

Day 90 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in General Science (7): National Quantum Mission's 2026 Progress

READING MATERIAL

Covers the National Quantum Mission's framework and its concrete 2026 progress milestones.

Q: When was the National Quantum Mission (NQM) approved, and what is its total budget/timeline?

A: Approved by the Union Cabinet on 19 April 2023, with a total budget of ₹6,003.65 crore, running from 2023-24 to 2030-31.

Q: What are 2 of the NQM's key technical targets?

A: Developing intermediate-scale quantum computers with 50-1,000 physical qubits within 8 years; and satellite-based secure quantum communication between ground stations over a range of 2,000 km within India.

Q: What quantum communication milestone was announced in 2026?

A: A historic 1,000-km secure quantum communication milestone, achieved in LESS THAN 2 YEARS.

Q: What is the Quantum Valley project (Amaravati), and what is its 2026 target?

A: A dedicated quantum-technology hub in Amaravati, aiming to establish India's first 133-qubit quantum computer by 2026.

Q: How many Thematic Hubs (T-Hubs) has the NQM established, and what do they focus on?

A: 4 Thematic Hubs, focusing on: Quantum Computing, Quantum Communication, Quantum Sensing & Metrology, and Quantum Materials & Devices.

Q: What startup and education support has the NQM provided so far?

A: 8 startups supported under NQM's guidelines; 23 academic institutions approved to establish quantum teaching laboratories (with 100+ more proposals under evaluation).

MOCK TEST (WITH ANSWERS)

1. The National Quantum Mission was approved by the Union Cabinet on:

A. 19 April 2022

B. 19 April 2023 ✓

C. 19 April 2024

D. 19 April 2025

Explanation: 19 April 2023. [hard, web-research]

2. The National Quantum Mission's total budget is approximately:

A. ₹3,000 crore

B. ₹4,500 crore

C. ₹6,000 crore ✓

D. ₹8,000 crore

Explanation: ₹6,003.65 crore. [hard, web-research]

3. In 2026, the National Quantum Mission achieved a secure quantum communication milestone of:

A. 500 km

B. 1,000 km ✓

C. 1,500 km

D. 2,000 km

Explanation: 1,000 km. [medium, web-research]

4. The Quantum Valley project at Amaravati aims to establish India's first quantum computer with how many qubits, by 2026?

A. 50 qubits

B. 100 qubits

C. 133 qubits ✓

D. 250 qubits

Explanation: 133 qubits. [hard, web-research]

5. How many Thematic Hubs (T-Hubs) has the National Quantum Mission established?

- A. 2
- B. 3
- C. 4 ✓**
- D. 5

Explanation: 4 (Quantum Computing, Communication, Sensing & Metrology, Materials & Devices). [medium, web-research]

Part B – Aptitude & Mental Ability: Time and Work

READING MATERIAL

Time and Work is the last of the 10 named Aptitude skills — after this, Part B moves to its Reasoning unit. Real exam evidence shows heavy use of 'trap' questions designed to look like they need complex math but actually resolve via a simple ratio.

Q: Method — inverse proportion (more workers, less time): 12 men complete a work in 36 days. In how many days will 18 men finish the same work?

A: Men $\times$ Days = constant (total work). $12 \times 36 = 18 \times x \rightarrow x = 432/18 = 24$ days. More workers means proportionally fewer days.

Q: Method — sequential combined work: A, B, C can complete a job alone in 8, 12, 16 days respectively. A and C work together for 2 days, then C leaves and B joins A. In how many more days do A and B finish?

A: Work rates: A=1/8, B=1/12, C=1/16 per day. In 2 days, A+C complete $2 \times (1/8 + 1/16) = 2 \times (3/16) = 3/8$ of the work. Remaining = 5/8. A+B's combined rate = $1/8 + 1/12 = 5/24$ per day. Days needed = $(5/8) \div (5/24) = (5/8) \times (24/5) = 3$ days.

Q: Method — wages split by share of work done: A can do a job in 10 days, B in 15 days. They work together for 5 days, then C finishes the remaining work in 2 days. Total payment for the whole job is Rs.5,000 — how should it be split?

A: Pay is split in proportion to how much of the TOTAL work each person actually did, not how many days they worked. In 5 days, A completes $5/10 = 1/2$ of the job, B completes $5/15 = 1/3$; together that's $5/6$, leaving $1/6$ for C to finish. Split of Rs.5,000 by work share: A gets $1/2 \times 5000 = \text{Rs.}2,500$; B gets $1/3 \times 5000 \approx \text{Rs.}1,667$; C gets $1/6 \times 5000 \approx \text{Rs.}833$. The key insight is the split follows WORK DONE, not days worked — C worked fewer days than B but did a comparable or different share depending on the rates, and pay tracks output, not time.

Q: Method — pairwise combined rates $\rightarrow$ solve for all three together: A+B finish a job in 12 days, B+C in 15 days, C+A in 20 days. In how many days can A, B, C finish it together?

A: Add all three pairwise rates: $(A+B) + (B+C) + (C+A) = 1/12 + 1/15 + 1/20 = 2(A+B+C)$'s rate. LCM=60: $5/60 + 4/60 + 3/60 = 12/60 = 1/5$. So $2 \times (A+B+C \text{ rate}) = 1/5 \rightarrow (A+B+C) \text{ rate} = 1/10 \rightarrow$ together they finish in 10 days.

Q: Trap question: If 9 spiders make 9 webs in 9 days, how many days does 1 spider take to make 1 web?

A: 9 days — NOT 1 day. Each spider makes 1 web in 9 days regardless of how many spiders work in parallel (they're not collaborating on the SAME web). This tests whether you recognize the rate-per-individual stays fixed, the classic trap being to assume the answer scales down to 1.

Q: Method — pipes filling and draining together: Two pipes fill a tank in 30 and 40 minutes separately; a third pipe drains it in 24 minutes. With all three open, how long to fill the tank?

A: Combined rate = $1/30 + 1/40 - 1/24$ (subtract the draining pipe's rate). LCM=120: $4/120 + 3/120 - 5/120 = 2/120 = 1/60$. So the tank fills in 60 minutes.

Q: Method — provisions lasting fewer days with more people (inverse proportion again): 1,000 soldiers have 70 days of provisions. If 400 more soldiers join, how long will provisions last?

A: People $\times$ Days = constant (total provisions). $1000 \times 70 = 1400 \times x \rightarrow x = 70000/1400 = 50$ days.

Q: Method — find the second worker's solo time from a combined time and the first worker's solo time: A and B together finish a job in 16 days; A alone takes 48 days. How long does B alone take?

A: B's rate = $(A+B)$'s rate - A's rate = $1/16 - 1/48 = 3/48 - 1/48 = 2/48 = 1/24$. So B alone takes 24 days.

Q: Trap question: If 5 persons complete 5 projects in 5 days, how long do 50 persons take to complete 50 projects?

A: Still 5 days — NOT 50 days. The ratio of people to projects stays the same (1:1), so the per-project time doesn't change; 50 people working on 50 projects is just 10 independent copies of the original 5-person/5-project scenario happening in parallel.

MOCK TEST (WITH ANSWERS)

1. 15 men complete a work in 24 days. In how many days will 20 men finish the same work?

- A. 16 days
- B. 18 days ✓**
- C. 20 days
- D. 30 days

Explanation: $15 \times 24 = 20 \times x \rightarrow x = 360/20 = 18$ days. [easy, authored]

2. A and B together can finish a job in 18 days; A alone takes 45 days. How long does B alone take?

- A. 24 days

- B. 27 days
- C. 30 days ✓**
- D. 36 days

Explanation: B's rate = $\frac{1}{18} - \frac{1}{45} = \frac{5}{90} - \frac{2}{90} = \frac{3}{90} = \frac{1}{30}$. B alone takes 30 days. [medium, authored]

3. If 7 workers build 7 walls in 7 days, how many days does 1 worker take to build 1 wall?

- A. 1 day
- B. 7 days ✓**
- C. 14 days
- D. 49 days

Explanation: 7 days — each worker's individual rate is unaffected by how many work in parallel on separate walls. [medium, authored]

4. Two pipes fill a tank in 12 and 15 minutes separately; a third pipe drains it in 20 minutes. With all three open, how long to fill the tank?

- A. 8 minutes
- B. 10 minutes ✓**
- C. 12 minutes
- D. 15 minutes

Explanation: Rate = $\frac{1}{12} + \frac{1}{15} - \frac{1}{20}$. LCM=60: $\frac{5}{60} + \frac{4}{60} - \frac{3}{60} = \frac{6}{60} = \frac{1}{10}$. Time = 10 minutes. [medium, authored]

5. 800 soldiers have provisions for 60 days. If 200 more soldiers join, how long will the provisions last?

- A. 40 days
- B. 45 days
- C. 48 days ✓**
- D. 50 days

Explanation: $800 \times 60 = 1000 \times x \rightarrow x = 48000/1000 = 48$ days. [easy, authored]

6. A, B, and C can do a job alone in 24, 30, and 40 days respectively. Working together, how long will they take?

- A. 8 days
- B. 10 days ✓**
- C. 12 days
- D. 15 days

Explanation: Combined rate = $\frac{1}{24} + \frac{1}{30} + \frac{1}{40}$. LCM=120: $\frac{5}{120} + \frac{4}{120} + \frac{3}{120} = \frac{12}{120} = \frac{1}{10}$. Together they take 10 days. [medium, authored]

7. If 8 people complete 8 tasks in 8 days, how long do 40 people take to complete 40 tasks?

- A. 8 days ✓**
- B. 20 days
- C. 40 days
- D. 320 days

Explanation: Still 8 days — the people-to-task ratio (1:1) is unchanged, so per-task time is unchanged. [medium, authored]

Part C – General English: Poem: "Wandering Singers" — Sarojini Naidu

READING MATERIAL

A Sarojini Naidu poem about folk musicians who travel from place to place, detached from worldly ties, singing of history, beauty, and fate. Real coaching-site evidence tests theme, in-context meaning, and figure-of-speech together.

Q: What is the central theme of "Wandering Singers"?

A: A wandering life and detachment — the singers move freely from place to place (compared to birds, butterflies, and clouds), unattached to any single home, spreading songs of love, beauty, history, and fate.

Q: What do the singers' songs cover, thematically?

A: Glorious cities whose past has been forgotten; the past lives of noble, happy, and beautiful women; and mighty battles fought by brave kings — in short, beauty, history, glory, bravery, happiness, and sorrow.

Q: What does "The voice of the wind is the voice of our fate" mean, and what figure of speech is it?

A: It means they follow their destiny wherever it leads — like the wind, their path isn't fixed or self-chosen. It's a Metaphor (fate is directly equated with the wind's voice, not compared using 'like/as').

Q: What does "All men are our kindred, the world is our home" imply about the singers?

A: They feel at home anywhere — they treat all people as family and don't see themselves as tied to one fixed home.

Q: What instrument do the wandering singers carry?

A: A lute.

Q: Who wrote "Wandering Singers"?

A: Sarojini Naidu.

MOCK TEST (WITH ANSWERS)

1. "Wandering Singers" was written by:

- A. Rabindranath Tagore
- B. Sarojini Naidu ✓**
- C. Sarala Devi
- D. Subramania Bharati

Explanation: Sarojini Naidu. [easy, web-research]

2. The central theme of the poem is:

- A. Modern music
- B. Freedom of travel
- C. Sacrifice and patriotism
- D. Wandering life and detachment ✓**

Explanation: Wandering life and detachment. [medium, web-research]

3. "The voice of the wind is the voice of our fate" means:

- A. Wind can be controlled
- B. They control their destiny
- C. They follow destiny wherever it leads ✓**
- D. They fear the wind

Explanation: They follow destiny wherever it leads, just as the wind's direction isn't chosen. [medium, web-research]

4. "The voice of the wind is the voice of our fate" uses which figure of speech?

- A. Simile
- B. Personification
- C. Metaphor ✓**
- D. Alliteration

Explanation: Metaphor — fate is directly equated with the wind's voice. [medium, web-research]

5. "All men are our kindred, the world is our home" implies that the singers:

A. Live in houses

B. Feel at home anywhere ✓

C. Dislike strangers

D. Stay only in forests

Explanation: They feel at home anywhere, treating all people as family. [medium, web-research]

6. What instrument do the wandering singers carry?

A. Harp

B. Lute ✓

C. Drum

D. Flute

Explanation: A lute. [easy, web-research]